STensor/Variation (Experimental)

Setup

In[1]:=
```
<< mGRG`STensor`
```

In[2]:=
```
$PreRead = ReplaceAll[#,
    expr_String :> StringReplace[expr, "SP" → "mGRG`STensor`Private"]] &;
```

In[3]:=
```
DefKind[Greek, Alphabet["Greek"]];
DefKind[Capital, ToUpperCase /@ Alphabet[]];
DefKind[Bar, "b" <> # & /@ Alphabet[]];
DefKind[Dot, "d" <> # & /@ Alphabet[]];
DefKind[Hat, "h" <> # & /@ Alphabet[]];
```

In[8]:=
```
Tdefine[f[]]; Tdefine[v[ua], 1]; Tdefine[T, 2]; Tdefine[F, "-ba"];
```

In[9]:=
```
SetAttributes[{c1, c2}, Constant]
```

Variational Derivative

DumFresh

In[10]:=
```
DumFresh::usage
```

Out[10]=
```
DumFresh[expr] renames all dummy indices in 'expr' to new unique indices.
```

In[11]:=
```
ResetDummies::usage
```

Out[11]=
```
ResetDummies[expr, opts] renames dummy indices in 'expr' to
  a canonical form (e.g., la, lb, ...), avoiding conflicts with
  existing free indices. Options like HeadQs and IndexQs can be
  used to control which parts of the expression are processed.
```

In[12]:=
```
ResetDummiesFlag::usage
```

Out[12]=
```
A flag that controls whether dummy indices in an expression
  are automatically reset to a canonical form in the output.
```

In[13]:=
```
Off[ResetDummiesFlag]
```

In[14]:=
```
VD[f[], T[la, lb] × v[ua]]
```
Out[14]=

$\left[v^{\text{Latin\$10331}}\left(\delta_f T_{\text{Latin\$10331b}}\right)\right] + \left[T_{\text{Latin\$10331b}}\left(\delta_f v^{\text{Latin\$10331}}\right)\right]$

In[15]:=
```
On[ResetDummiesFlag]
```

In[16]:=
```
VD::usage
```
Out[16]=

VD[arg, expr] represents the
variational derivative of 'expr' with respect to 'arg'.

In[17]:=
```
VD[f[], T[la, lb] × v[ua]]
```
Out[17]=

$\left[v^a\left(\delta_f T_{ab}\right)\right] + \left[T_{ab}\left(\delta_f v^a\right)\right]$

Linear, Leibnitz rule, and Derivation

In[18]:=
```
VD[f[], c1 T[la, lb] × v[ua] + c2 v[la]]
```
Out[18]=

$c1\left[v^a\left(\delta_f T_{ab}\right)\right] + c2\left[\left(\delta_f v_a\right)\right] + c1\left[T_{ab}\left(\delta_f v^a\right)\right]$

Scalar Functions

In[19]:=
```
VD[Metricg[la, lb], Log[f[]]]
```
Out[19]=

$\left[\frac{1}{f}\left(\delta_{g_{ab}} f\right)\right]$

In[20]:=
```
VD[Metricg[la, lb], Log[v[la] × v[ua]]]
```
Out[20]=

$\left[\frac{v^c}{v_a\ v^a}\left(\delta_{g_{ab}} v_c\right)\right] + \left[\frac{v_c}{v_a\ v^a}\left(\delta_{g_{ab}} v^c\right)\right]$

In[21]:=
```
VD[Metricg[la, lb], (v[la] × v[ua])^n]
```
Out[21]=

$\left[\text{Log}\left[v_{\text{Latin\$10398}}\ v^{\text{Latin\$10398}}\right]\ \left(v_a\ v^a\right)^n\left(\delta_{g_{ab}} n\right)\right] + \left[n\ \left(v_a\ v^a\right)^{-1+n}\ v^c\left(\delta_{g_{ab}} v_c\right)\right] + \left[n\ v_c\ \left(v_a\ v^a\right)^{-1+n}\left(\delta_{g_{ab}} v^c\right)\right]$

In[22]:=
```
VD[Metricg[la, lb], Tscalar[v[la] × v[ua]]]
```
Out[22]=

$\left[v^c\left(\delta_{g_{ab}} v_c\right)\right] + \left[v_c\left(\delta_{g_{ab}} v^c\right)\right]$

In[23]:=
```
% // FullForm
```
Out[23]//FullForm=
```
Plus[VD[Metricg[la, lb], v[lc], v[uc]], VD[Metricg[la, lb], v[uc], v[lc]]]
```

Variations by a Tensor

Symmetric metric:

In[24]:=
```
VD[Metricg[la, lb], Metricg[lc, ld]]
```
Out[24]=
$$\frac{1}{2}\,\delta^{a}{}_{d}\,\delta^{b}{}_{c}+\frac{1}{2}\,\delta^{a}{}_{c}\,\delta^{b}{}_{d}$$

In[25]:=
```
VD[Metricg[la, lb], Metricg[uc, ud]]
```
Out[25]=
$$-\frac{1}{2}\,g^{ad}\,g^{bc}-\frac{1}{2}\,g^{ac}\,g^{bd}$$

Anti - symmetric tensor :

In[26]:=
```
VD[F[la, lb], F[lc, ld]]
```
Out[26]=
$$-\frac{1}{2}\,\delta^{a}{}_{d}\,\delta^{b}{}_{c}+\frac{1}{2}\,\delta^{a}{}_{c}\,\delta^{b}{}_{d}$$

In[27]:=
```
VD[F[la, lb], F[uc, ud]]
```
Out[27]=
$$-\frac{1}{2}\,g^{ad}\,g^{bc}+\frac{1}{2}\,g^{ac}\,g^{bd}$$

In[28]:=
```
Tdefine[R, "abcd+bcad+cabd"]
```

In[29]:=
```
VD[R[la, lb, lc, ld], R[le, lf, lg, lh]]
```
Out[29]=
$$\frac{1}{3}\,\delta^{a}{}_{f}\,\delta^{b}{}_{g}\,\delta^{c}{}_{e}\,\delta^{d}{}_{h}+\frac{1}{3}\,\delta^{a}{}_{g}\,\delta^{b}{}_{e}\,\delta^{c}{}_{f}\,\delta^{d}{}_{h}+\frac{1}{3}\,\delta^{a}{}_{e}\,\delta^{b}{}_{f}\,\delta^{c}{}_{g}\,\delta^{d}{}_{h}$$

In[30]:=
```
VD[R[la, lb, lc, ld], R[le, uf, lg, uh]]
```
Out[30]=
$$\frac{1}{3}\,\delta^{b}{}_{g}\,\delta^{c}{}_{e}\,g^{af}\,g^{dh}+\frac{1}{3}\,\delta^{a}{}_{e}\,\delta^{c}{}_{g}\,g^{bf}\,g^{dh}+\frac{1}{3}\,\delta^{a}{}_{g}\,\delta^{b}{}_{e}\,g^{cf}\,g^{dh}$$

In[31]:=
```
VD[R[la, lb, lc, ld], R[le, uμ, lg, lh]]
```

Msg: Different kinds of indices: ub and u μ

Out[31]=

$$\frac{1}{3}\,\delta^{b}{}_{g}\,\delta^{c}{}_{e}\,\delta^{d}{}_{h}\,\delta^{a\mu}+\frac{1}{3}\,\delta^{a}{}_{e}\,\delta^{c}{}_{g}\,\delta^{d}{}_{h}\,\delta^{b\mu}+\frac{1}{3}\,\delta^{a}{}_{g}\,\delta^{b}{}_{e}\,\delta^{d}{}_{h}\,\delta^{c\mu}$$

In[32]:=
```
VD[R[la, lb, lc, ld], R[le, lf, lg]]
```

Msg: Different lengths of indices: 4 and 3

Out[32]=

R_{efg}

In[33]:=
```
Off[MetricgFlag]
```

In[34]:=
```
VD[R[la, lb, lc, ld], R[le, lf, lg, lh]]
```

Out[34]=

$$\frac{1}{3}\,\delta^{a}{}_{f}\,\delta^{b}{}_{g}\,\delta^{c}{}_{e}\,\delta^{d}{}_{h}+\frac{1}{3}\,\delta^{a}{}_{g}\,\delta^{b}{}_{e}\,\delta^{c}{}_{f}\,\delta^{d}{}_{h}+\frac{1}{3}\,\delta^{a}{}_{e}\,\delta^{b}{}_{f}\,\delta^{c}{}_{g}\,\delta^{d}{}_{h}$$

In[35]:=
```
VD[R[la, lb, lc, ld], R[le, uf, lg, lh]]
```

Msg: No metric for Latin

Out[35]=

$$\frac{1}{3}\,\delta^{b}{}_{g}\,\delta^{c}{}_{e}\,\delta^{d}{}_{h}\,\delta^{af}+\frac{1}{3}\,\delta^{a}{}_{e}\,\delta^{c}{}_{g}\,\delta^{d}{}_{h}\,\delta^{bf}+\frac{1}{3}\,\delta^{a}{}_{g}\,\delta^{b}{}_{e}\,\delta^{d}{}_{h}\,\delta^{cf}$$

In[36]:=
```
On[MetricgFlag]
```

In[37]:=
```
Tdefine[t[la, lμ]]
```

In[38]:=
```
VD[t[la, lb], t[uc, ud]] (* No Greek's metric *)
```

Msg: No metric for Greek

Out[38]=

$g^{ac}\,\delta^{bd}$

In[39]:=
```
VD[t[la, lb], t[uc, uμ]]
```

Msg: Different kinds of indices: ub and u μ

Out[39]=

$g^{ac}\,\delta^{b\mu}$

In[40]:=
```
VD[t[la, lν], t[uc, uμ]] (* No Greek's metric *)
```

Msg: No metric for Greek

Out[40]=

$g^{ac}\,\delta^{\nu\mu}$

IndependentVD

In[41]:=
```
IndependentVD::usage
```
Out[41]=
```
An option for VD that specifies a list of tensors to be considered
   independent of the variation. The variational derivative of an
   independent tensor with respect to another tensor will be zero.
```

In[42]:=
```
VD[f[], T[la, lb] × v[ua]]
```
Out[42]=

$[v^a(\delta_f T_{ab})] + [T_{ab}(\delta_f v^a)]$

In[43]:=
```
VD[f[], T[la, lb] × v[ua], IndependentVD → {T}]  (* δfT==0 *)
```
Out[43]=

$[T_{ab}(\delta_f v^a)]$

Integration by Parts

In[44]:=
```
ByPartsVD::usage
```
Out[44]=
```
An option for VD that, when set to True, enables
   integration by parts for specified derivative operators.
```

$$\int \delta_{v_a}[(\partial_b v_c)(\partial^b v^c)] = \int (\partial_b v_c)\, \delta_{v_a}(\partial^b v^c) + \int (\partial^b v^c)\, \delta_{v_a}(\partial_b v_c)$$
$$= -\int (\partial^b \partial_b v_c)\, g^{ac} - \int (\partial_b \partial^b v^c)\, \delta^a_c$$

In[45]:=
```
BD[lb, v[lc]] × BD[ub, v[uc]]
```
Out[45]=

$\partial_a v_b\, \partial^a v^b$

In[46]:=
```
VD[v[la], %, ByPartsVD → True]
```
Out[46]=

$-\partial_b \partial^b v^a - \partial^b \partial_b v_c\, g^{ac}$

Implementation formulas:

$$\int (\partial^b v^c)\, \delta_{v_a}(\partial_b v_c) = -\int (\partial_b \partial^b v^c)\, \delta^a_c$$

In[47]:=
```
VD[v[la], BD[lb, v[lc]], BD[ub, v[uc]], ByPartsVD → True]
```
Out[47]=

$-\partial_b \partial^b v^a$

$$\int (\partial_b v_c)\, \delta_{v_a}(\partial^b v^c) = -\int (\partial^b \partial_b v_c)\, g^{ac}$$

In[48]:=
```
VD[v[la], BD[ub, v[uc]], BD[lb, v[lc]], ByPartsVD → True]
```

Out[48]=

$-\partial^{b}\partial_{b}v_{c}\ g^{ac}$